

Alibaba B2B E-commerce Database Management System

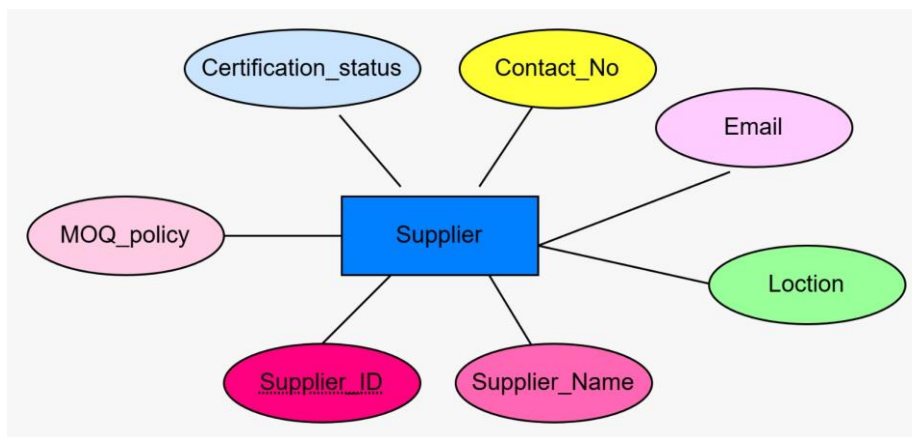
Chapter 3

3.1 Introduction

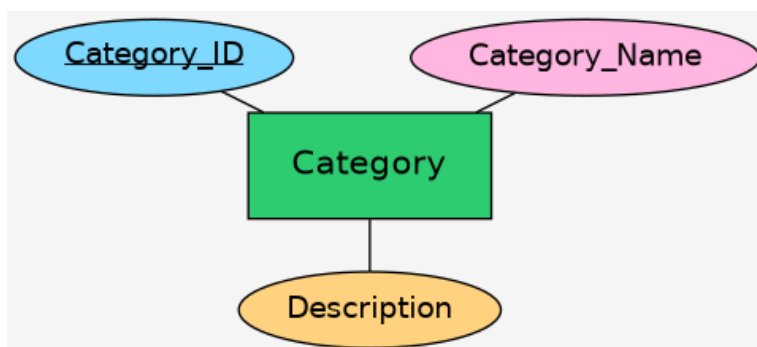
The Entity-Relationship (ER) Diagram for the Alibaba B2B E-Commerce Database Management System provides a conceptual representation of the database structure designed around the JBL Bluetooth earbuds wholesale sourcing case study. It identifies the main entities, their attributes, primary keys, and the relationships between them. The relationships are defined using appropriate cardinality and participation constraints to represent the interactions among Suppliers, Products, Customers, Orders, Order_Items, Payments, Categories, and Reviews. The ER diagram serves as the foundation for designing the relational database schema and implementing the database efficiently.

3.2 Entity Relationship (ER) Diagram and Attributes

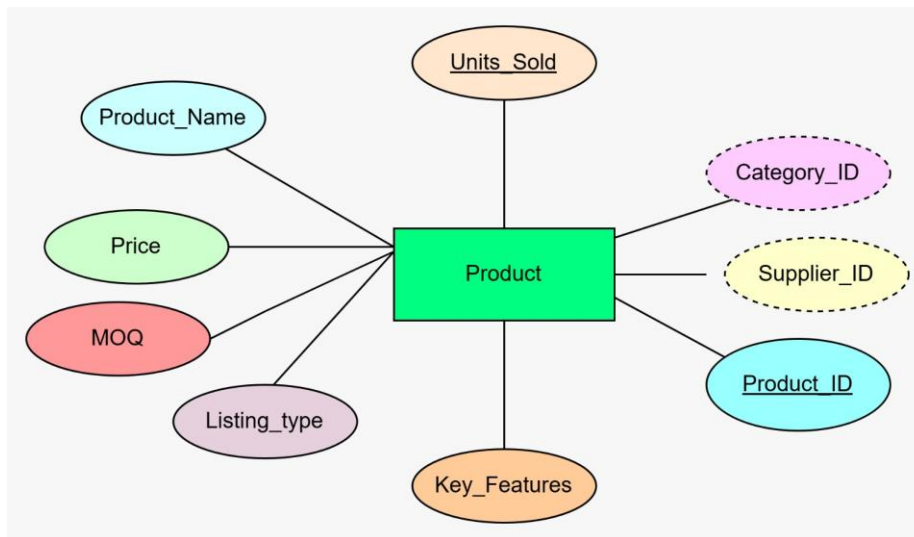
i. Supplier



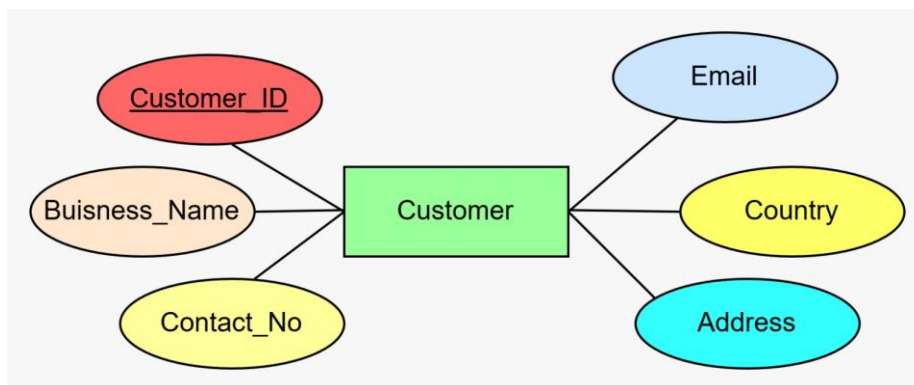
ii. Category



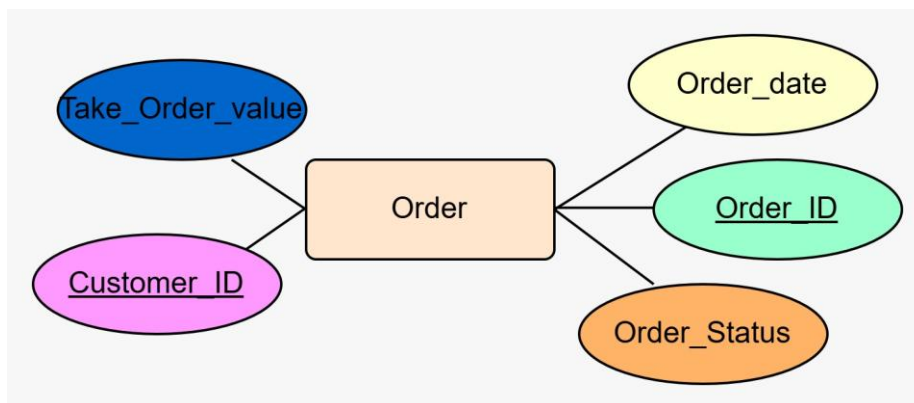
iii. Product



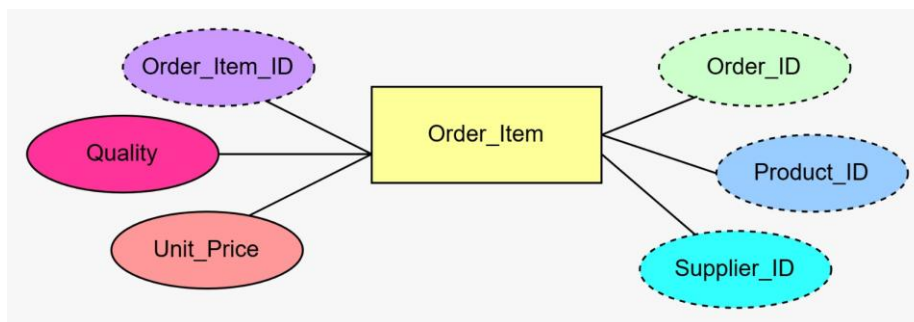
iv. Customer



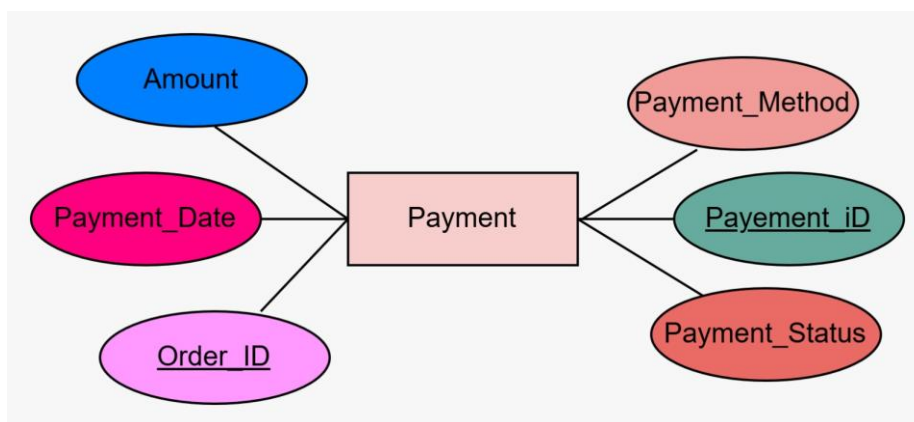
v. Order



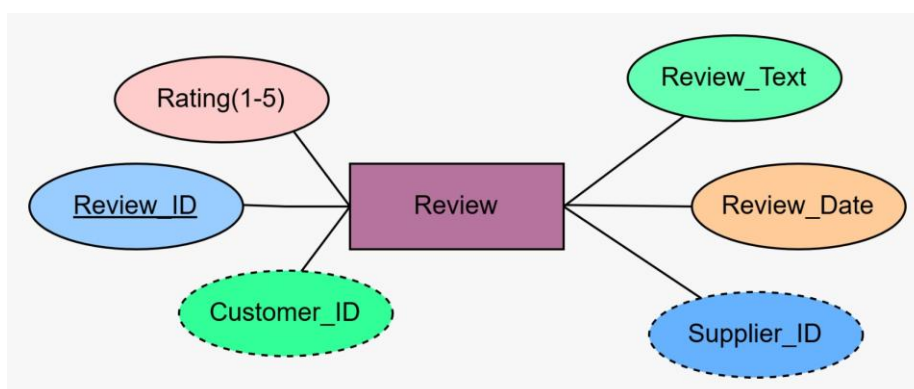
vi. Order_Item



vii. Payment

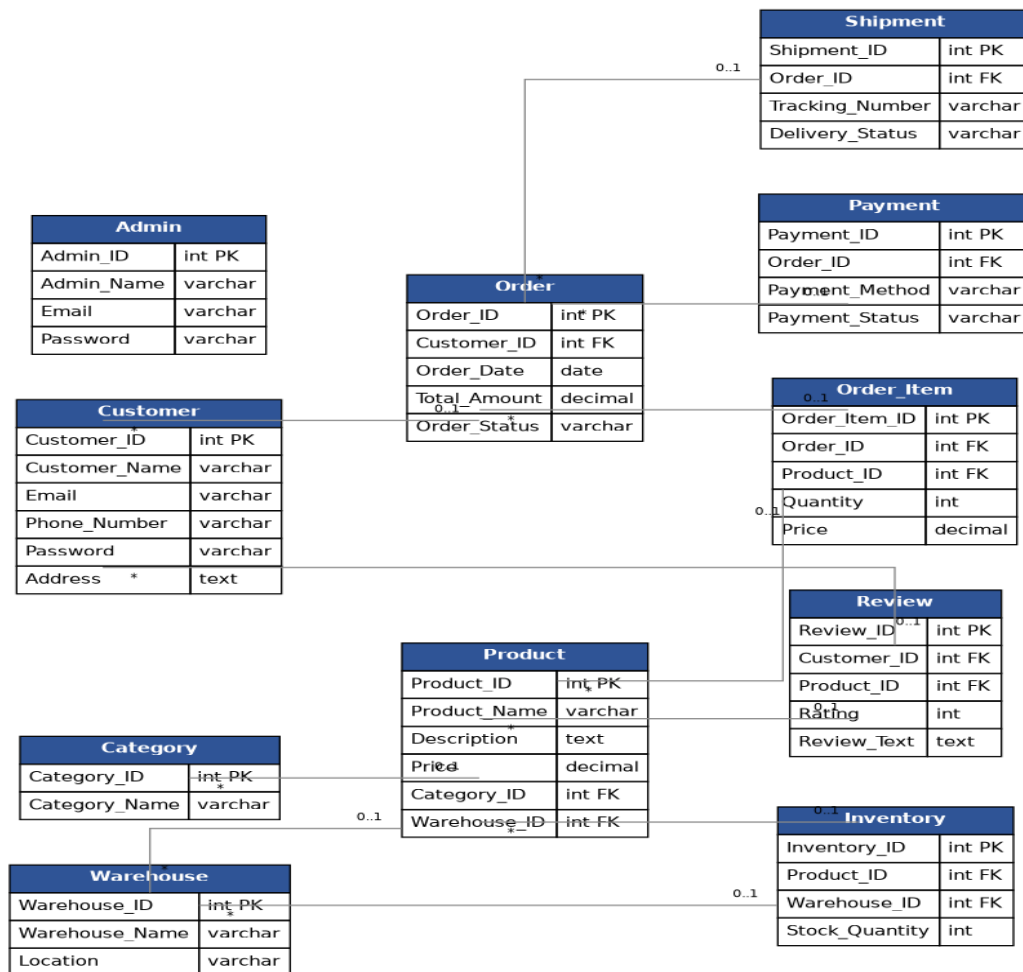


viii. Review



3.3 Relational Schema

The relational schema diagram below shows the tables, columns, and foreign-key relationships derived from the entities described in Section 3.2.



3.4 Overall ER Diagram

The table below summarizes the relationships between entities, along with their cardinality and participation constraints, followed by the consolidated Entity-Relationship diagram for the Alibaba B2B E-commerce Database Management System.

Entity 1	Relation	Entity 2	Cardinality	Participation
Supplier	Lists	Product	One-to-Many	Partial / Total
Category	Categorizes	Product	One-to-Many	Total / Partial
Customer	Places	Order	One-to-Many	Partial / Total
Order	Contains	Order_Item	One-to-Many	Total / Total
Product	Included-in	Order_Item	One-to-Many	Partial / Total
Supplier	Fulfills	Order_Item	One-to-Many	Partial / Total

Order	Generates	Payment	One-to-One	Total / Total
Customer	Writes	Review	One-to-Many	Partial / Total
Supplier	Receives	Review	One-to-Many	Partial / Total

